

Vodacom M-Pesa Tanzania

Customer-to-Business (C2B) API Documentation

Open API – Single Stage Payment

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1. Overview

The **Customer-to-Business (C2B) Single Stage** API allows a business to request payment from a customer's M-Pesa wallet. When the request is submitted, M-Pesa sends a **USSD Push** prompt to the customer's phone. The customer enters their M-Pesa PIN to authorize the payment. Funds are then transferred from the customer's wallet to the business's M-Pesa short code / service provider account.

Typical use cases: e-commerce checkout, bill payments, service payments, ticket purchases, and any online/offline collection of funds.

2. Prerequisites

- Registered business account with Vodacom Tanzania M-Pesa Business (short code / Service Provider Code).
- Developer account on the M-Pesa Open API Portal: <https://openapiportal.m-pesa.com>
- API Key and Public Key obtained after creating an application and enabling **C2B Single Payment**.
- Sandbox Service Provider Code for testing: **000000** (or the one shown in your portal).
- Sandbox test MSISDN: **0000000000001**.

3. Authentication Flow

All API calls require a **Session Key**. The Session Key is obtained by encrypting your API Key with the Public Key (RSA) and calling the Get Session endpoint.

3.1 Get Session Key

Endpoint

GET <https://openapi.m-pesa.com/sandbox/ipg/v2/vodacomTZN/getSession/>
GET <https://openapi.m-pesa.com/openapi/ipg/v2/vodacomTZN/getSession/> (Production)

Headers

Header	Value	Required
Authorization	Bearer {Encrypted_API_Key}	Yes
Origin	developer.mpesa.vm.co.mz (or as shown in portal)	Yes
Content-Type	application/json	Yes

Note: Encrypt your raw API Key using the RSA Public Key (PEM) provided in the portal (usually OAEP with SHA-256). The resulting Base64 string is used as the Bearer token for the Get Session call. The returned Session Key is then used as the Bearer token for all subsequent API calls (valid for the lifetime configured in your application, typically 24 hours).

4. C2B Single Stage Endpoint

Request

POST https://openapi.m-pesa.com/sandbox/ipg/v2/vodacomTZN/c2bPayment/singleStage/
POST https://openapi.m-pesa.com/openapi/ipg/v2/vodacomTZN/c2bPayment/singleStage/ (Production)

Headers

Header	Value	Required
Authorization	Bearer {Session_Key}	Yes
Content-Type	application/json	Yes
Origin	As configured / portal value	Yes

Request Body Parameters

Parameter	Type	Required	Description / Example
input_Amount	String	Yes	Amount to charge (whole number). Example: "1000"
input_CustomerMSISDN	String	Yes	Customer phone in international format (255...). Sandbox: "000000000001"
input_Country	String	Yes	Always "TZN" for Tanzania
input_Currency	String	Yes	Always "TZS" for Tanzania
input_ServiceProviderCode	String	Yes	Your business short code. Sandbox: "000000"
input_TransactionReference	String	Yes	Your unique transaction reference (max ~20 chars). Example: "INV-20260922-001"
input_ThirdPartyConversationID	String	Yes	Unique ID generated by your system for tracking. Example: UUID
input_PurchasedItemsDesc	String	Yes	Short description of what is being paid for. Example: "Order #4521 - Shoes"

Sample Request Body (JSON)

```
{
  "input_Amount": "1500",
  "input_CustomerMSISDN": "255712345678",
  "input_Country": "TZN",
  "input_Currency": "TZS",
  "input_ServiceProviderCode": "123456",
  "input_TransactionReference": "ORD-98765",
  "input_ThirdPartyConversationID": "a1b2c3d4-e5f6-7890-abcd-ef1234567890",
  "input_PurchasedItemsDesc": "Payment for Order #98765"
}
```

5. Responses

5.1 Successful Synchronous Response

When the request is accepted by the gateway and the USSD Push is successfully initiated, you receive an immediate synchronous response:

```
{
  "output_ResponseCode": "INS-0",
  "output_ResponseDesc": "Request processed successfully",
  "output_TransactionID": "RvvsqB0rcP3Y",
  "output_ConversationID": "1e029550d09745e7b2221bb4b2dc8ffc",
  "output_ThirdPartyConversationID": "a1b2c3d4-e5f6-7890-abcd-ef1234567890"
}
```

Response Fields

Field	Description
output_ResponseCode	Status code. "INS-0" means the request was accepted and USSD Push was sent.
output_ResponseDesc	Human-readable description of the response code.
output_TransactionID	M-Pesa platform transaction identifier (use for status queries / support).
output_ConversationID	Unique conversation ID generated by the Open API platform.
output_ThirdPartyConversationID	Echo of the ThirdPartyConversationID you sent in the request.

Important: A successful synchronous response (INS-0) only means the USSD Push was delivered. The actual payment is completed only after the customer enters the correct PIN. You should implement a callback / webhook (Response URL configured in the portal) or poll the Query Transaction Status API to confirm final success.

5.2 Asynchronous / Callback Response

If you configured a Response URL (webhook) in the Open API Portal application settings, M-Pesa will POST the final result to that URL once the customer completes or cancels the transaction.

```
{
  "input_OriginalConversationID": "1e029550d09745e7b2221bb4b2dc8ffc",
  "input_ThirdPartyReference": "a1b2c3d4-e5f6-7890-abcd-ef1234567890",
  "input_TransactionID": "RvvsqB0rcP3Y",
  "input_ResultCode": "0",
  "input_ResultDesc": "Request Processed Successfully"
}
```

6. Response & Error Codes

The Open API uses **INS-xxx** codes. Below are the most common codes you will encounter.

Code	Meaning	HTTP	Recommended Action
INS-0	Request processed successfully	200	Proceed – wait for customer PIN / callback
INS-1	Internal Error	500	Retry after a short delay; contact support if persistent
INS-2	Invalid API Key / Not enough balance	401	Check credentials or customer balance
INS-4	User is not active / Transaction failed	401	Verify customer account status
INS-5	Transaction cancelled by customer	401	Customer pressed Cancel – allow retry
INS-6	Transaction Failed / Not permitted	401	Show generic failure message
INS-9	Request timeout	408	Customer did not respond in time – retry
INS-10	Duplicate transaction	409	Use a new unique ThirdPartyConversationID / TransactionReference
INS-13	Invalid Shortcode Used	400	Verify ServiceProviderCode
INS-14	Invalid Reference Used	400	Check TransactionReference format/length
INS-15	Invalid Amount Used	400	Amount must be positive whole number
INS-16	Temporary overloading	503	Retry with exponential backoff
INS-17	Invalid Transaction Reference length	400	Keep between 1–20 characters
INS-19	Invalid ThirdPartyReference	400	Ensure unique and valid format
INS-20	Not All Parameters Provided	400	Check all required fields are present
INS-21	Parameter validations failed	400	Validate MSISDN, amount, country, currency
INS-26	Not authorized	400	Check Session Key and application permissions

INS-996	Customer Account Status Not Active	400	Customer must activate/register M-Pesa
INS-998	Invalid Market	400	Ensure Country=TZN and correct market endpoint

7. Best Practices & Recommendations

- Always generate a **unique** *input_ThirdPartyConversationID* (UUID recommended) and *input_TransactionReference* for every request to avoid INS-10 duplicates.
- Store the ConversationID and TransactionID returned in the synchronous response for reconciliation and support cases.
- Implement a secure HTTPS webhook (Response URL) and verify the source of callbacks.
- Handle timeouts and customer cancellations gracefully – present a clear “Retry Payment” option.
- Never hard-code production credentials; use environment variables / secret management.
- Test thoroughly in Sandbox using MSISDN **0000000000001** and Service Provider Code **000000** before requesting go-live.
- For production go-live, complete the full business onboarding process with Vodacom M-Pesa Business team (m-pesabusiness@vodacom.co.tz).

8. Useful Links

- Open API Portal: <https://openapiportal.m-pesa.com>
- <https://business.m-pesa.com/vodacom-tanzania/business-onboarding-tanzania/> Business Onboarding (Tanzania):
- Contact: m-pesabusiness@vodacom.co.tz

This document is a practical developer reference compiled from official Open API behaviour and community SDKs. Always cross-check the latest details inside the Open API Portal after logging in, as parameters and codes may be updated.

Document generated on 2026-09-22 • For integration support contact your Vodacom M-Pesa Business Manager